

José Neto

Senior Software Engineer — C++ | Python | 3D Graphics

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SUMMARY

Senior Software Engineer and Engineering Lead with an MSc in Computer Science (computer vision & 3D graphics) and 8+ years of experience spanning real-time rendering, 3D content pipelines, and backend systems. Currently leading development of a C++/Qt desktop application built on OpenUSD, Hydra, and MaterialX, and author of a commercial Unreal Engine 5 plugin. Comfortable owning complex systems end to end — from multithreaded GPU rendering internals to Python binding layers and build infrastructure — and known for pairing deep technical work with clear communication and mentorship.

WORK EXPERIENCE

Engineering Lead

10/2025 – Present

Darmstadt Graphics Group (DGG / RapidPipeline)

Remote

- Lead technical direction across development teams: architecture decisions, code reviews, and hands-on contributions to core C++ systems.
- Translate product vision into actionable technical plans; coordinate release planning and keep increments aligned with the roadmap.
- Act as the bridge between stakeholders and engineering, communicating changes, risks, and trade-offs.

Senior 3D Graphics Engineer

05/2022 – Present

Darmstadt Graphics Group (DGG / RapidPipeline)

Remote

- **Twin Studio (flagship desktop app):** core developer of a C++/Qt application for 3D model optimization and rendering, built on OpenUSD, Hydra, MaterialX, and the MoonRay production renderer.
- Designed multithreaded rendering features, including an offscreen 360° turntable render pipeline with a dedicated render thread, shared OpenGL contexts, and safe GPU resource lifecycle management.
- Built the application's Python scripting layer: embedded interpreter, in-app script editor/REPL, and a public API bridging two binding systems (Shiboken6 and nanobind) through a shared C++ pointer bridge.
- Implemented the product's licensing system end to end: browser-based activation with nonce polling, offline key files, and periodic revalidation.
- **RapidPipeline for Unreal:** lead developer of a commercial UE5 plugin for 3D asset optimization — Slate/UMG UI, static and skeletal mesh support, CAD import integration, and an action versioning system.
- **DCC pipeline tooling:** developed the DCC Importer (V-Ray/3ds Max materials to glTF 2.0) used to process hundreds of thousands of production assets; contributed to Blender plugin, cloud APIs, and a Babylon.js-based web viewer.

Stack: C++17, Qt, OpenUSD/Hydra, MaterialX, OpenGL, Unreal Engine 5, Python (Shiboken6, nanobind, PyTorch), CMake, TypeScript, Babylon.js

Backend Engineer & Tech Lead

10/2021 – 04/2022

Magnetis (investment fintech)

Remote

- Technical lead of the Broker team, responsible for the core accounting and financial operations platform: defined technical direction, coordinated development, and built backend services in Elixir/Phoenix.

Stack: Elixir, Phoenix, Ruby, Rails

Backend Engineer

01/2021 – 09/2021

First Security Technology AG

Remote

- Delivered key features for a network security management platform: 2FA login, PDF report generation, and message-based appliance control over RabbitMQ.
- Mentored junior developers on microservices architecture and domain logic.

Stack: PHP, Symfony, API Platform, RabbitMQ, SQL, C++

Software Analyst

08/2020 – 12/2020

Eldorado Institute (Motorola project)

Manaus, Brazil

- Native Android/C++ development on Motorola's camera engine as part of the night vision team.

Research Assistant

03/2020 – 08/2020

Universidade Federal de Minas Gerais (UFMG)

Belo Horizonte, Brazil

- Multimodal deep learning with Graph Convolutional Networks; co-authored a paper published in the *Computers & Graphics* journal.

Research Assistant
Universidade Federal do Rio Grande do Norte (UFRN)

09/2015 – 08/2019
Natal, Brazil

- Research on image and texture segmentation via deep learning; co-supervised an undergraduate whose capstone results were published at an international conference.

OPEN SOURCE

Contributor
cvc5 (SMT solver) & Carcara (Rust proof checker)

01/2024 – Present

- Brought cvc5 to the browser: WebAssembly/Emscripten builds and an interactive web tutorial running SMT-LIBv2 in-browser.
- Improved cross-language bindings (Python/Java), fixed a resource-limit infinite loop in the core solver, and extended CI configuration.
- Carcara: authored a procedural macro crate simplifying term-matching across the codebase, plus proof-rule test coverage.
- Built and operate a Django/Angular benchmarking platform with PAR-2 scoring, multi-source benchmark support, regression tracking, and automated reporting.

EDUCATION

MSc in Computer Science
Focus: Computer Vision & 3D Graphics

BSc in Computer Science
Minor in Geophysics

SKILLS

Core	C++ (14/17), Python, Rust, TypeScript, SQL
Graphics & 3D	OpenGL, OpenUSD, Hydra, MaterialX, Unreal Engine 5, glTF, Qt, Babylon.js
Tooling	CMake, Shiboken6, nanobind, PyTorch, RabbitMQ, Docker, CI/CD
Languages	Portuguese (native), English (fluent)